Collocation Dynamics in Physics-Informed Neural Networks

Adaptive sampling, diffusion, and solution sharpness

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Why study collocation dynamics?

The physics loss is evaluated on a finite set of points:

$$\mathcal{L}_{\mathrm{PDE}}(\theta; Z) = \frac{1}{N} \sum_{i=1}^{N} r_\theta(z_i)^2, \qquad z_i = (x_i, t_i).$$

- Point locations determine where the PDE is enforced during training.
- Keeping, replacing, or moving points changes coverage and the optimization trajectory, even at the same point budget.
- Concentrating points near difficult features may help, while leaving other regions insufficiently constrained.

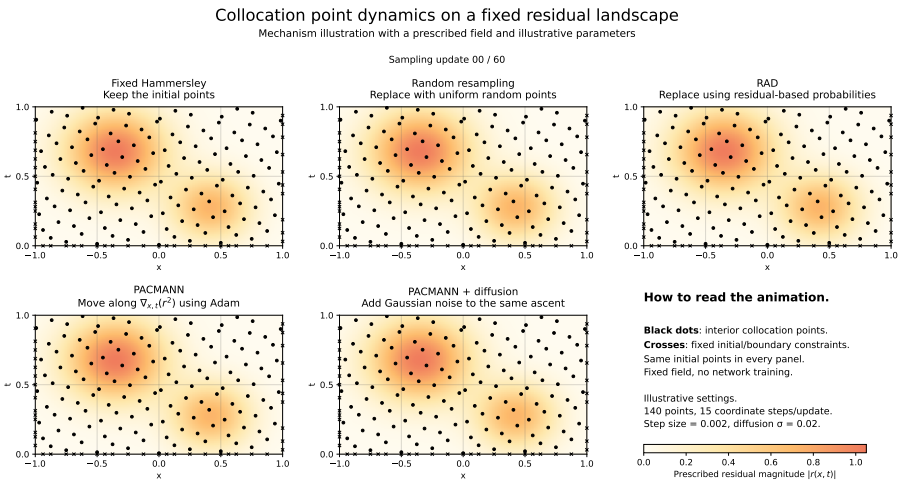
Five ways to place the same interior point budget

Method	Interior point strategy
Fixed Hammersley	Keep the initial low-discrepancy design.
Random resampling	Periodically replace points with fresh uniform random points.
RAD[1]	Sample fresh candidates preferentially where residual magnitude is large.
PACMANN[2]	Move points along the coordinate gradient of squared residual, using Adam ascent.
PACMANN + diffusion	Add domain-scaled Gaussian perturbations to the coordinate-Adam motion.

Residual-based selection versus gradient-based adaptive sampling

RAD selects a new population using residual scores; PACMANN moves the current population using $\nabla_{(x,t)} r_\theta^2$.

Collocation sampling: a mechanism illustration

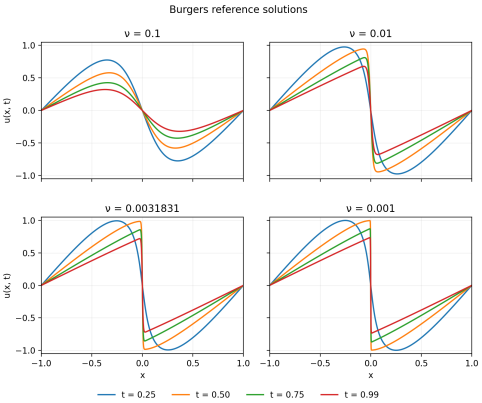


Starting problem: viscous Burgers equation

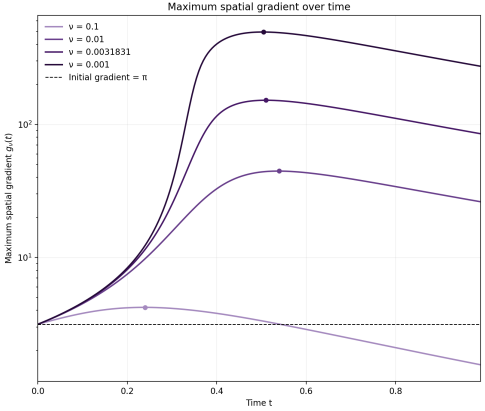
$$\partial_t u + u\,\partial_x u - \nu\,\partial_{xx} u = 0, \qquad (x,t) \in [-1,1] \times [0, 0.99].$$
$$u(x,0) = -\sin(\pi x), \qquad u(-1,t) = u(1,t) = 0.$$

- Viscosity ν controls diffusion in the PDE and the solution regime.
- Train from the PDE and fixed initial/boundary constraints.
- Use numerical reference solutions only for evaluation.

Reference solutions across viscosity regimes



Solution profiles. Colors identify the four physical times. The intermediate $\pi/100$ case is retained in the complete five-viscosity analysis.



Sharpness. Each curve tracks the steepest spatial slope over time; G is its largest value.

Research questions

1. How does the benefit of point-adaptive collocation change across solution-sharpness regimes, for solution and PDE-residual error?
2. Does adding diffusion to PACMANN improve either outcome, and does the effect depend on the regime?

Working hypothesis

Gradient-adaptive methods may offer greater benefits for solutions with sharp features. Both the adaptive-sampling and diffusion benefits remain open questions.

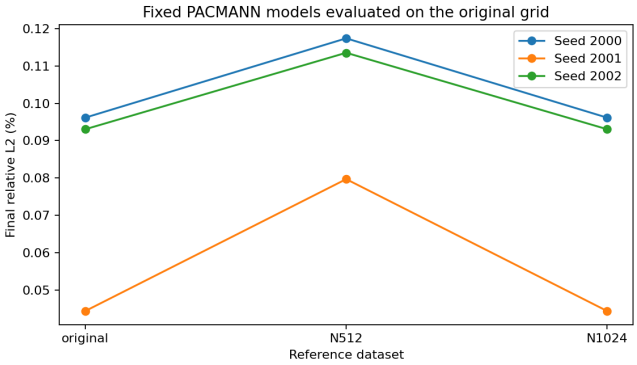
Reference generation and validation at $\nu = 0.01/\pi$

Generated reference

- Chebyshev spatial collocation, degree $N = 1024$.
- Radau time integration.
- Compare on the original 256 spatial points and 100 snapshot times.

Two different checks

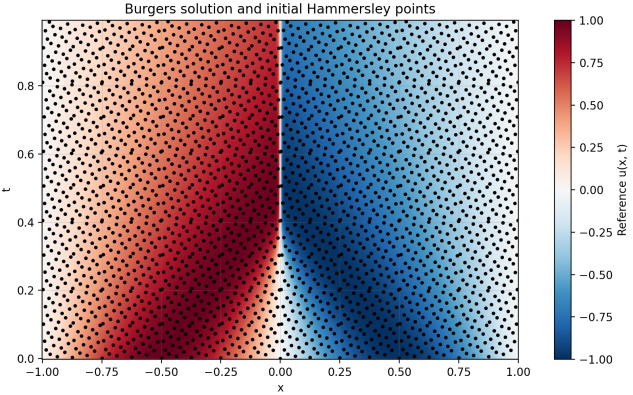
- Reference disagreement: 0.000319% relative L^2.
- N1024 change in fixed-model error scores: less than 0.006% relative.



Reading the plot. Each line is one fixed trained model. Only the reference used to score its predictions changes.

Interpretation. At this viscosity, the generated reference closely matches the original and gives essentially the same error scores for these three trained models.

Matched training and independent evaluation



Training controls

- Same ten paired seeds and initial weights per method.
- 2,500 PDE; 80 BC; 160 IC.
- Interior updates during Adam only.

Final-model evaluation

- Solution: uniform 1024×199 grid.
- Residual RMS: independent 1024×200 midpoint grid.
- Boundary / initial RMSE.

Initial design. Shared points at $\nu = 0.01/\pi$.

Comparison principle. Every method is evaluated on the same fixed grids, regardless of how it places its training points.

Detailed scientific settings

Setting	Frozen configuration
Network / precision	2–64–64–64–64–1, tanh, Glorot normal; float64.
Point budgets	2,500 interior PDE points; 80 boundary and 160 initial points.
Optimizer schedule	Five phases, each with 7,000 Adam steps at 10^{-3}, followed by L-BFGS.
L-BFGS limits	Nominal maxiter 3,000, maxfun 3,750; maxcor 100, gtol 10^{-8}, ftol 0.
Sampling cadence	Every 50 Adam steps; no point updates during L-BFGS.
PACMANN motion	15 coordinate steps at 10^{-5}; $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-7}$ outside the square root.
RAD	100,000 candidates; weight $\lvert r\rvert / \operatorname{mean}(\lvert r\rvert) + 1$; sample without replacement.
Diffusion convention	Per coordinate step, add independent Gaussian noise with componentwise standard deviation $\sigma(L_x, L_t)/\sqrt{15}$.
Evaluation	Uniform solution grid 1024×199 with spatial barycentric reference interpolation; independent residual grid 1024×200.

Tune first, then freeze settings

σ tuning: three paired seeds

- Seeds 3000–3002.
- Start with $\sigma = 0$ and positive values $10^{-5}, 10^{-4}, 10^{-3}, 10^{-2}$.
- Expand positive boundary winners (at most two rounds).
- Select by mean final solution relative L^2 error.

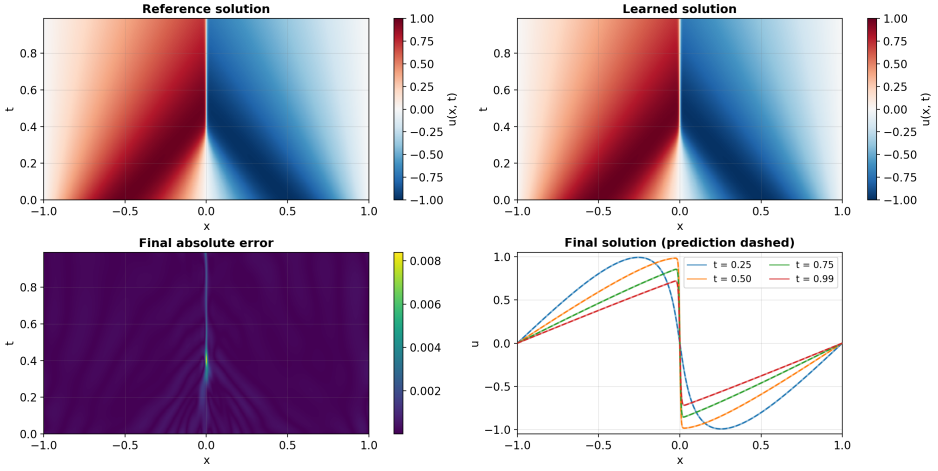
Evaluation: ten paired seeds

- Freeze σ before using seeds 4000–4009.
- Compare all five methods; retain all seeds.

Displayed viscosity ν	0.1	0.01	$0.01/\pi$	0.001
Frozen diffusion σ	10^{-5}	10^{-3}	10^{-4}	10^{-6}

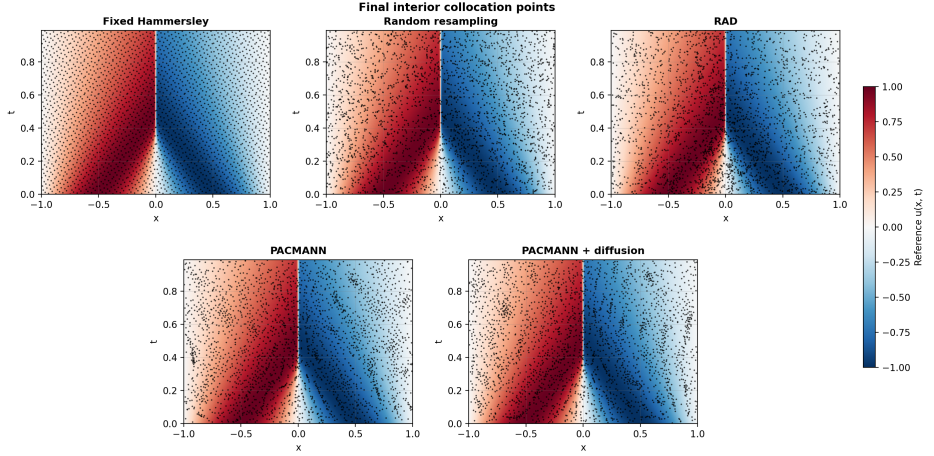
Tuning scope. Diffusion is tuned separately at each viscosity; the other samplers retain their upstream parameter settings.

PACMANN solution example at $\nu = 0.01/\pi$



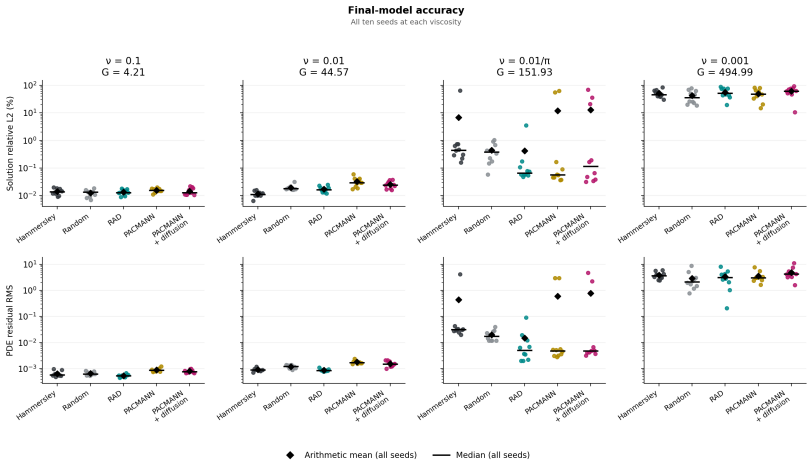
Illustrative successful run. First planned seed (4000), final solution relative $L^2 = 0.0454\%$. The method comparisons retain all ten seeds.

Final collocation positions across methods



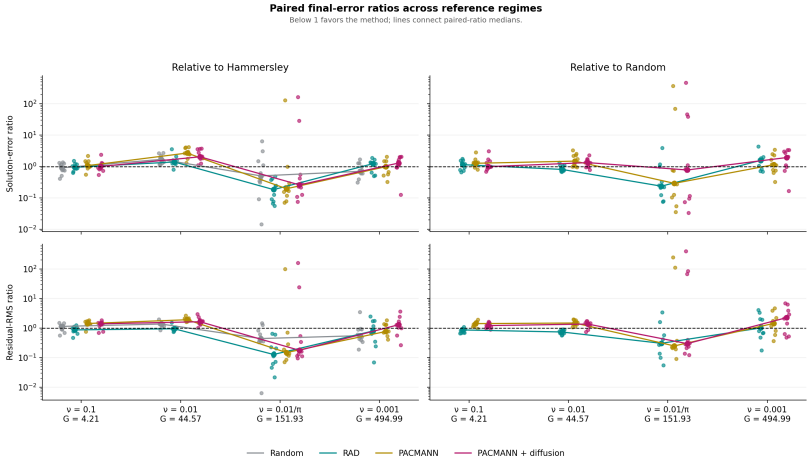
Shared reference background; black dots are the 2,500 final interior points ($\nu = 0.01/\pi$, seed 4000).

Final accuracy and seed variability



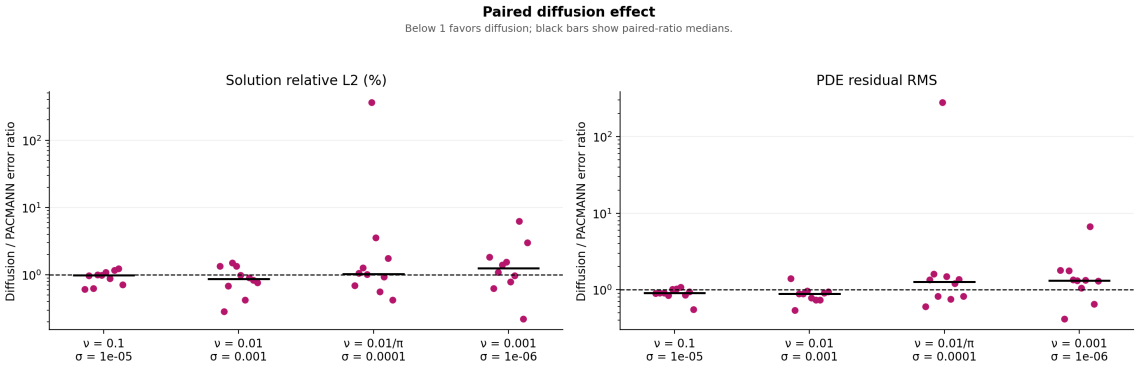
At $\nu = 0.01/\pi$, low PACMANN median errors coexist with very large errors in some seeds. At $\nu = 0.001$, both means and medians are large for every method under the shared training budget.

Paired benefits relative to both baselines



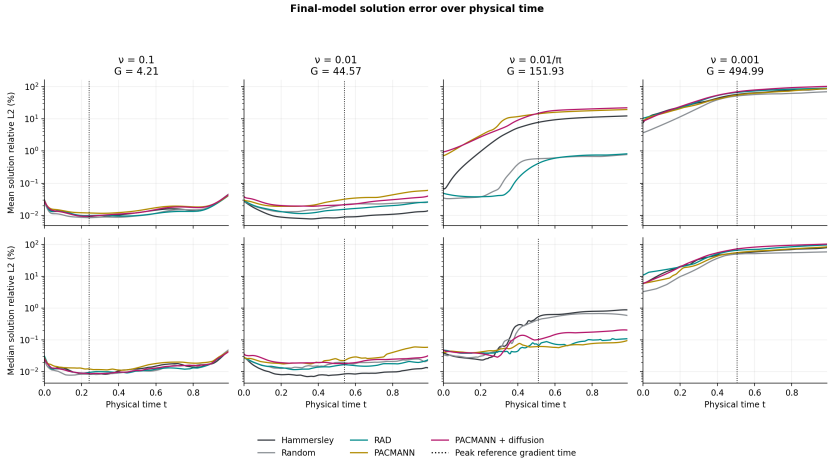
RAD and PACMANN have favorable median ratios against both baselines at $\nu = 0.01/\pi$. This pattern does not extend to $\nu = 0.001$ against random resampling.

Does diffusion improve PACMANN?



From left to right, diffusion lowers solution error in **7/10**, **7/10**, **4/10**, and **4/10** pairs; residual RMS in **7/10**, **9/10**, **4/10**, and **2/10**. Both mean errors decrease at $\nu = 0.1$ and 0.01, and increase at $0.01/\pi$ and 0.001.

Where in physical time do final-model errors grow?



At $\nu = 0.01/\pi$, large-error runs raise the later-time mean above the median. At $\nu = 0.001$, both are large, including before the peak reference gradient. Dotted lines mark that peak time, not front onset.

Conclusions, limitations, and next steps

- **Adaptive collocation:** At $\nu = 0.01/\pi$, PACMANN has lower median solution error than Hammersley and random resampling after the sharp front forms. Large-error seeds limit robustness; no consistent advantage across regimes. At $\nu = 0.001$, all methods have large errors.
- **Diffusion:** Lower mean errors than ordinary PACMANN at $\nu = 0.1$ and 0.01; higher at $0.01/\pi$ and 0.001. No general benefit in the displayed regimes.
- **Limits and design choices:** One PDE, ten seeds per setting, a fixed training budget, reference checks that vary by viscosity, and per-viscosity diffusion tuning.
- **Next steps:** Develop a σ controller driven by local solution gradients. Extend to 1D Allen–Cahn with a varying interface-width parameter, 2D Burgers, and other problems.

References

[1] C. Wu, M. Zhu, Q. Tan, Y. Kartha, and L. Lu.
A comprehensive study of non-adaptive and residual-based adaptive sampling for physics-informed neural networks.
2023.

[2] C. Visser, A. Heinlein, and B. Giovanardi.
PACMANN: Point adaptive collocation method for artificial neural networks.
2025.

[3] L. Lu, X. Meng, Z. Mao, and G. E. Karniadakis.
DeepXDE: A deep learning library for solving differential equations.
SIAM Review, 2021.

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Appendix: PACMANN implementation changes

Category	Implementation choice
Retained	Four 64-neuron tanh layers, float64, five alternating Adam/L-BFGS stages, sampler settings, coordinate-Adam procedure, keep initial/boundary constraints fixed.
Changed	Match 2,500 interior PDE points from initialization for every method; Use generated references and common final-checkpoint evaluation.
Added / tuned	Gaussian diffusion of point coordinates; select its coefficient separately at each viscosity on tuning seeds.

- DeepXDE[3] with a PyTorch backend, adapted from the released implementation.
- Evaluate final models and retain all planned seeds, including failures.
- Record actual optimizer work; shared limits do not imply equal computation.

Appendix: reference checks, tuning, and computation

Reference comparison at $\nu = 0.01/\pi$

- Compare at the original 256 spatial points and 100 snapshot times.
- Use barycentric interpolation in space at exact shared times.
- Score the same three trained models without retraining.

Diffusion selection details

- At $\nu = 0.01/\pi$, zero was eligible for selection.
- At later viscosities, select the positive sigma with the lowest mean final solution error; zero is a comparison baseline.
- A positive boundary winner triggers search expansion, with at most two expansion rounds.

Computational setup

Numerics	Float64.
Software	DeepXDE 1.14.0; PyTorch 2.7.1.
Hardware	RTX 5070, 12 GB VRAM.
Execution	Ten concurrent seeds per method; method batches run sequentially.

Sharper reference at $\nu = 0.001$

- Degree $N = 4096$; other references use $N = 1024$.
- N2048–N4096 disagreement: 0.00965% relative L^2.
- A baseline grid-refinement check supported keeping the common evaluation grids. These checks do not bound absolute reference error.